

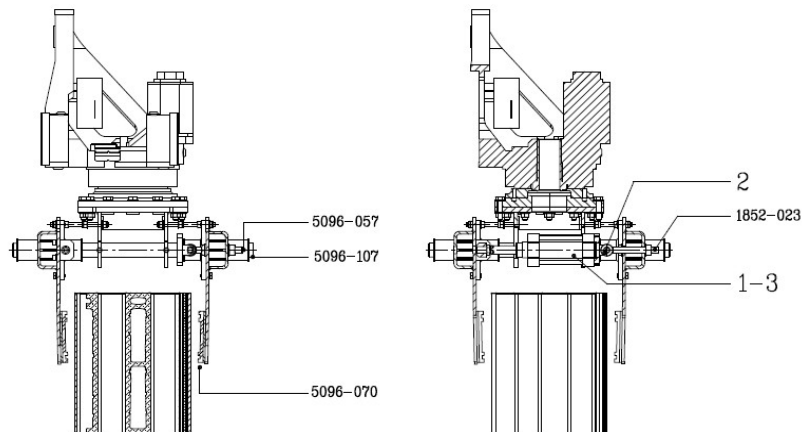
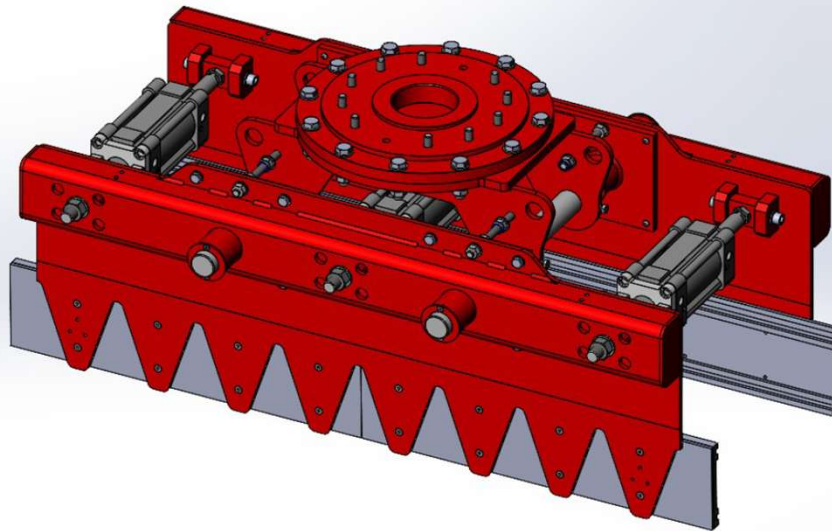
Robot with Transfer Clamp of large Packs over the Strapping Line -Relationship SOPs

MACHINE TYPE: PK-120 SERIAL NUMBER :M.02240

FP McCANN (Cadeby)

ORDER : C22004

YEAR OF PRODUCTION 2023

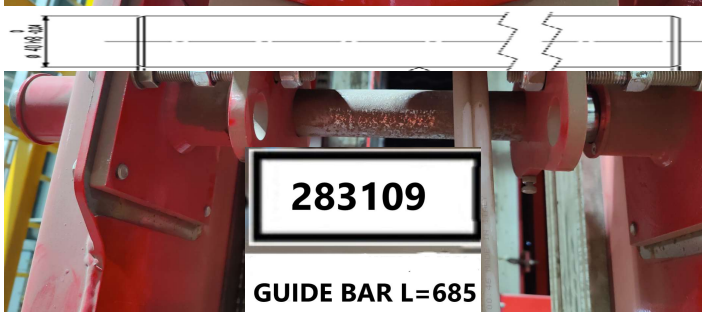


SOP M.02240.00 List Robot with Transfer Clamp of large Packs over the Strapping Line -Relationship SOPs
SOP M.02240.01 Rubber Plate and Guide Bar
SOP M.02240.02 Pneumatic and Electrical Components for Gripper Robot

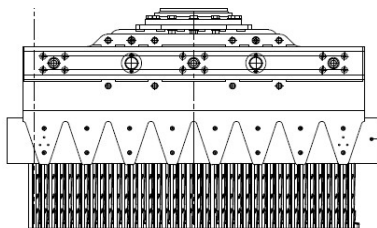
Robot with Transfer Clamp of large Packs over the Strapping Line

MACHINE TYPE: PK-120 SERIAL NUMBER :M.02240

Rubber Plate and Guide Bar



267826

VULCANIZED RUBBER
PLATE L=600

267826

VULCANIZED RUBBER
PLATE L=600General Description and Function

*This robotic clamp is responsible for securely gripping and transferring large tile bundles over the strapping line.

*The primary wear component is the vulcanized rubber plate that makes direct contact with the concrete tile surfaces.

Critical Points and Risk Management

*Associated with rubber plate wear is the potential degradation of the two guide bars that allow the allow the rubber jaw plates to move.

*Wear on these guide bars can lead to improper clamping force and bundle instability during transfer

SUPERVISION LEVEL**DAILY**Clamp Performance:

*Observe the clamp during bundle pickup and transfer.

*Identify any bundle slippage, misalignment, or irregular movement during pickup and transport to the vertical strapping station.

Vulcanized Rubber Plate Wear Assessment:

* Check the degree of deterioration on all rubber contact plates to determine if replacement is needed.

Guide Bar Wear & Play:

*Inspect the guide bars for wear, scoring, or excessive clearance. Check the amount of torsional play allowed at the extreme ends of the rubber jaw plates.

Clamp Positioning & Alignment:

*Verify that the open/close positions of the clamp jaws are consistently maintained by the 4 threaded rods and adjustment nuts, ensuring they remain in their preset positions.

WEEKLY

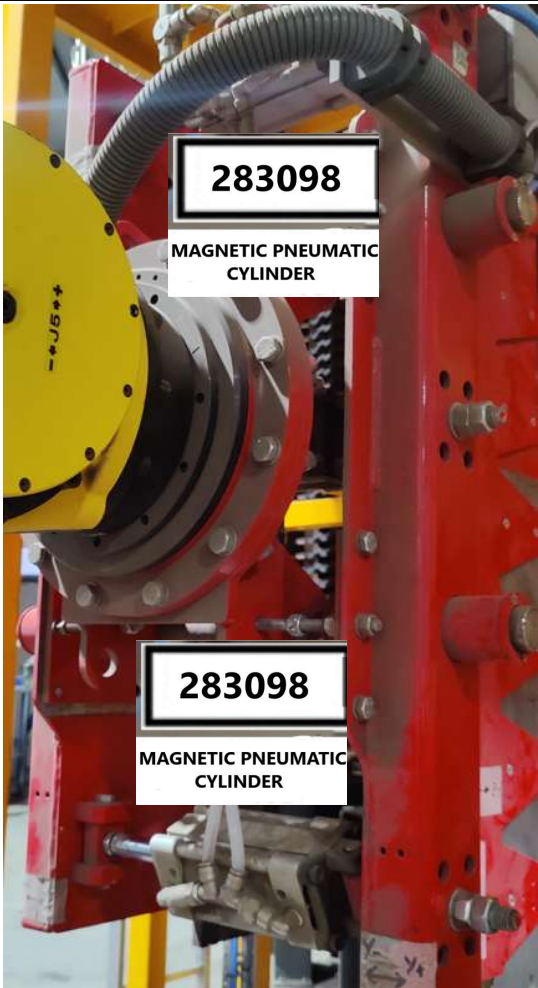
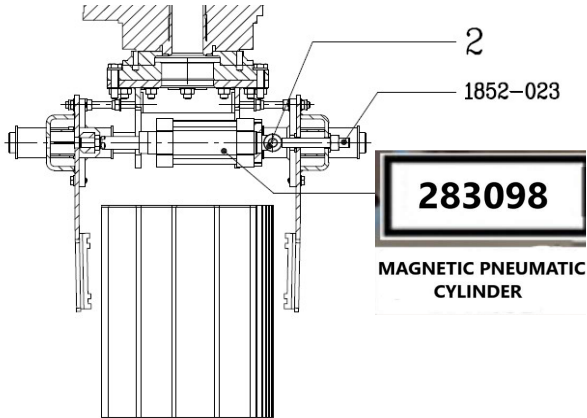
267826 Vulcanized rubber plate L=600 (4 units)

283109 Guide bar L=685 (2 units)

CLEANING
WEEKLY

*Clamp structure surfaces, rubber plates, and guide bar sliding surfaces

*Use appropriate tools (scrapers, brushes, dry cloths) to remove all accumulated concrete dust, debris, and residual material.

VORTEX HYDRA	STANDARD OPERATING PROCEDURE	Code: SOP M.02240.02 15/12/2025 Page: 3 de 3
Robot with Transfer Clamp of large Packs over the Strapping Line		
MACHINE TYPE: PK-120 SERIAL NUMBER :M.02240		
Pneumatic and Electrical Components for Gripper Robot		
  		General Description and Function *The gripper unit is responsible for securely picking up and transferring tile bundles to the strapping station. Its pneumatic and electrical components are critical for reliable operation. Critical Points and Risk Management *Air leaks or improper pressure regulation in the supply to the three cylinders can cause bundle release during transfer, leading to product damage and line stoppage. SUPERVISION LEVEL WEEKLY 283098 Magnetic pneumatic cylinder 276852 Magnetic sensor 277178 Solenoid valve MONTHLY 37195 Tie-rod for pneumatic cylinder Ø80 266618 Guide bar support + #2 Bushings (it. 17561) 276849 Rear female hinge for pneumatic cylinder DAILY Air Leak Detection: *Listen and look for air leaks at cylinders, tubes, and fittings during gripper open/close maneuvers. *Verify that the air supply pressure regulator maintains the correct setting. Regulator Marking: *The working pressure must be clearly marked on the regulator gauge to enable quick visual confirmation WEEKLY External Magnetic Sensors: *Verify the tightness of all sensor mountings. *The reference position on the cylinder must be clearly marked to detect any displacement. Solenoid Valve Connections: *Inspect electrical connectors for security and integrity. CLEANING WEEKLY *Cylinders surfaces, sensors and guide bars. *Use appropriate tools (scrapers, brushes, dry cloths) to remove all accumulated debris, dust, and residual material from surfaces